SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI – 602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CASE STUDY

Course Code:	DSA0502	**Course Title:**	Query Processing for Data Science
CO Assessed:	CO4	**Assessment:**	Tool 1– CASE STUDY
Weightage:		**Total Marks:**	
CO4 Statement: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.			
Student Name:	BHARANI DHARAN S	**Reg. No.:**	192424141
Section / Batch:	D / 2024	**Date of Submission:**	17/08/26
Faculty In-charge:	**Dr. B.Shyamala Gowri**	**Project Mode:**	Individual Submission

Code of Conduct

I BHARANI DHARAN S{192424141} (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

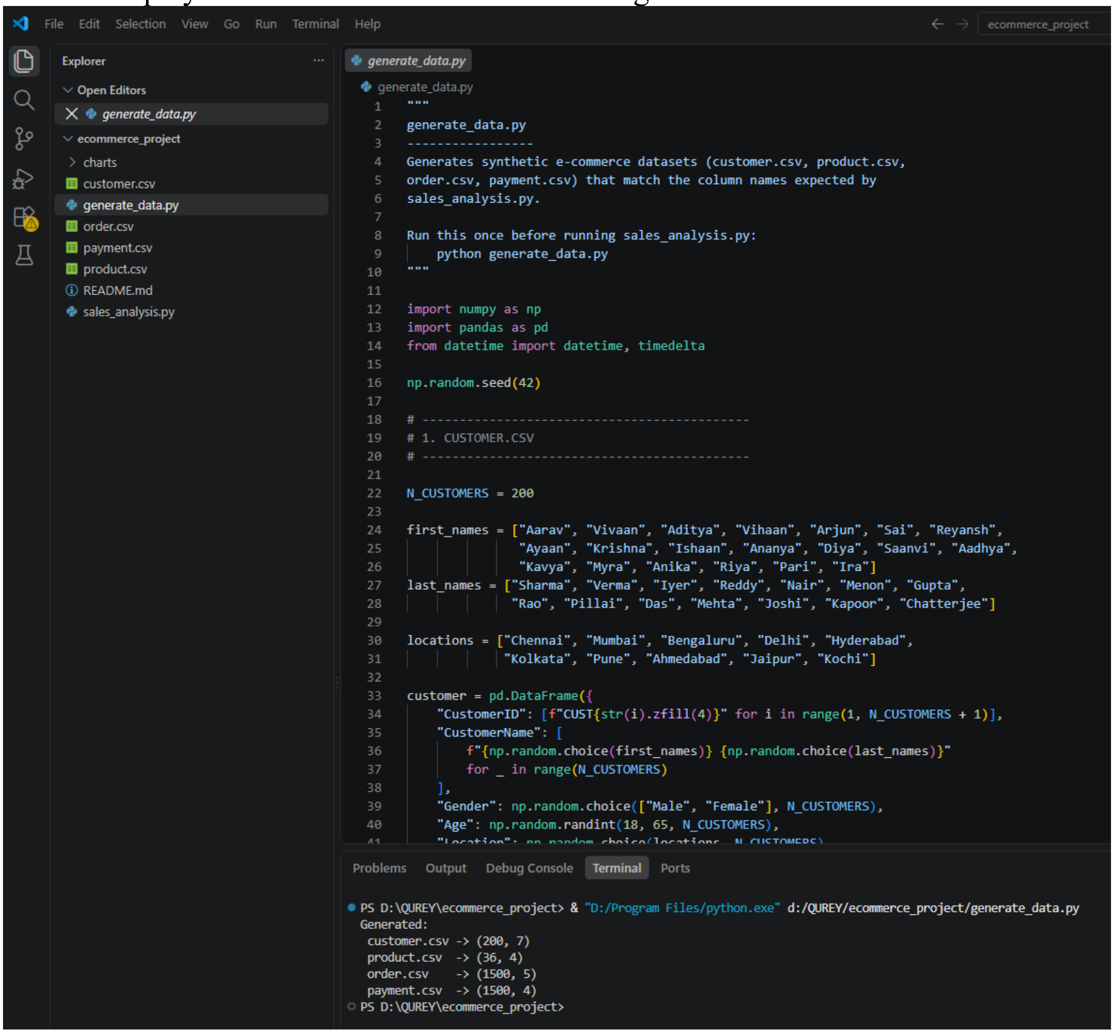
Signature: BHARANI DHARAN S

Aim

To generate synthetic e-commerce datasets containing customer, product, order, and payment information using Python, so that the datasets can be used for e-commerce sales analysis.

Algorithm

1. Import the required Python libraries: NumPy, Pandas, and datetime.
2. Set a random seed to generate reproducible data.
3. Generate 200 customer records with customer ID, name, gender, age, location, email, and signup date.
4. Create product records with product ID, product name, category, and price.
5. Generate 1500 order records by randomly selecting customers and products, along with quantity and order date.
6. Insert a few extreme quantity values to create outliers for later analysis.
7. Generate payment records containing order ID, payment method, payment status, and transaction ID.
8. Convert all generated data into Pandas DataFrames.
9. Save the DataFrames as customer.csv, product.csv, order.csv, and payment.csv.
10. Display the names and dimensions of the generated datasets.

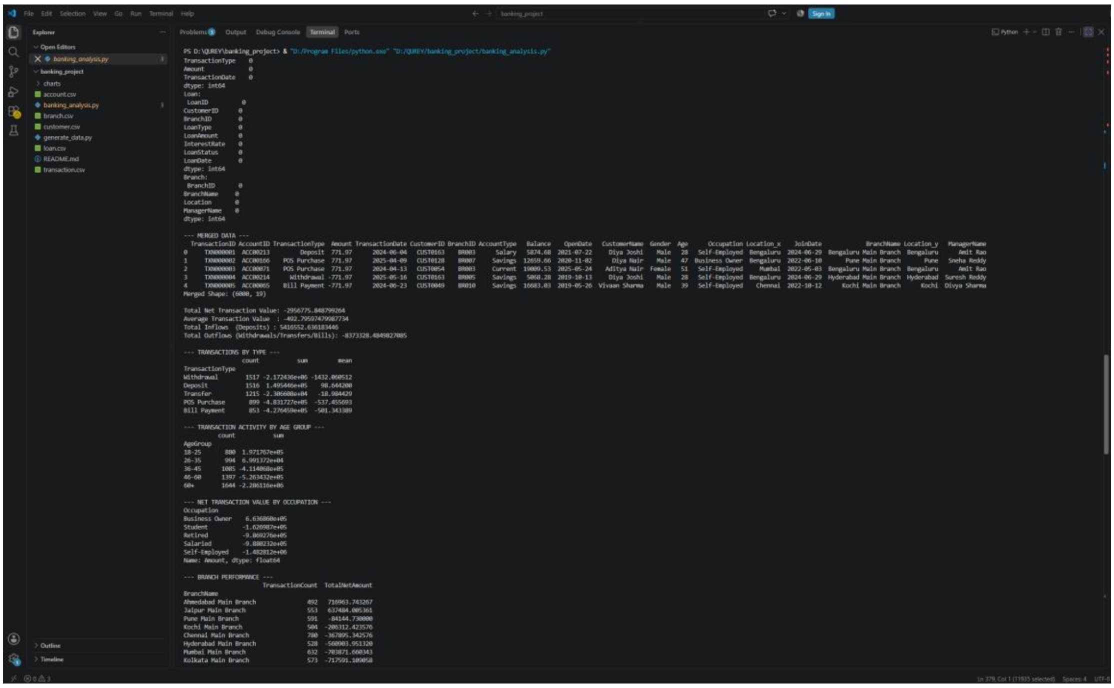


AIM

To perform **Banking Transaction Analysis** using Python by integrating customer, account, transaction, loan, and branch datasets, and to analyze transaction patterns, customer behavior, branch performance, loan information, trends, correlations, and unusual transactions using suitable statistical methods and visualizations.

ALGORITHM

1. **Start**
2. Import required Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and OS.
3. Load the following CSV datasets:
    - Customer
    - Account
    - Transaction
    - Loan
    - Branch
4. Display the dataset shape, sample records, data types, and missing values.
5. Remove duplicate records from all datasets.
6. Merge the datasets using common keys such as CustomerID, AccountID, and BranchID.
7. Calculate:
    - Total transaction value
    - Average transaction value
    - Total inflow
    - Total outflow
8. Group transactions by **Transaction Type** and analyze transaction count and amount.
9. Categorize customers into **age groups** and analyze their transaction activity.
10. Analyze transaction values based on **occupation**.
11. Evaluate **branch performance** using transaction volume and net transaction value.
12. Analyze loans based on **loan type, loan amount, and loan status**.
13. Convert transaction dates into monthly periods and analyze **monthly transaction trends**.
14. Generate a **correlation matrix and heatmap** for numerical variables.
15. Detect unusual transactions using the **IQR (Interquartile Range) method**.
16. Generate a histogram to analyze the distribution of transaction amounts.
17. Analyze account balances and calculate the **average balance by account type**.
18. Create suitable charts for all major analyses and save them in the charts folder.
19. Identify key findings and provide recommendations based on the analysis.
20. **Stop.**


File Edit Selection View Go Run Terminal Help
banking_project
Sign in
Explorer
Open Editors
banking_analysis.py
banking_project
charts
account.csv
banking_analysis.py
branch.csv
customer.csv
generate_data.py
loan.csv
README.md
transaction.csv
Outline
Timeline
Problems Output Debug Console Terminal Ports
PS D:\QUREY\banking_project> & "D:/Program Files/python.exe" "D:/QUREY/banking_project/banking_analysis.py"
TransactionType 0
Amount 0
TransactionDate 0
dtype: int64
Loan:
LoanID 0
CustomerID 0
BranchID 0
LoanType 0
LoanAmount 0
InterestRate 0
LoanStatus 0
LoanDate 0
dtype: int64
Branch:
BranchID 0
BranchName 0
Location 0
ManagerName 0
dtype: int64
--- MERGED DATA ---
TransactionID AccountID TransactionType Amount TransactionDate CustomerID BranchID AccountType Balance OpenDate CustomerName Gender Age Occupation Location_x JoinDate BranchName Location_y ManagerName
0 TXN000001 ACC00213 Deposit 771.97 2024-06-04 CUST0163 BR003 Salary 5874.68 2021-07-22 Diya Joshi Male 28 Self-Employed Bengaluru 2024-06-29 Bengaluru Main Branch Bengaluru Amit Rao
1 TXN000002 ACC00166 POS Purchase 771.97 2025-04-09 CUST0128 BR007 Savings 12659.66 2020-11-02 Diya Nair Male 47 Business Owner Bengaluru 2022-06-10 Pune Main Branch Pune Sneha Reddy
2 TXN000003 ACC00071 POS Purchase 771.97 2024-04-13 CUST0054 BR003 Current 19009.53 2025-05-24 Aditya Nair Female 51 Self-Employed Mumbai 2022-05-03 Bengaluru Main Branch Bengaluru Amit Rao
3 TXN000004 ACC00214 Withdrawal -771.97 2025-05-16 CUST0163 BR005 Savings 5068.28 2019-10-13 Diya Joshi Male 28 Self-Employed Bengaluru 2024-06-29 Hyderabad Main Branch Hyderabad Suresh Reddy
4 TXN000005 ACC00065 Bill Payment -771.97 2024-06-23 CUST0049 BR010 Savings 16683.03 2019-05-26 Vivaan Sharma Male 39 Self-Employed Chennai 2022-10-12 Kochi Main Branch Kochi Divya Sharma
Merged Shape: (6000, 19)
Total Net Transaction Value: -2956775.848799264
Average Transaction Value : -492.79597479987734
Total Inflows (Deposits) : 5416552.636183446
Total Outflows (Withdrawals/Transfers/Bills): -8373328.4849827085
--- TRANSACTIONS BY TYPE ---
count sum mean
TransactionType
Withdrawal 1517 -2.172436e+06 -1432.060512
Deposit 1516 1.495446e+05 98.644200
Transfer 1215 -2.306608e+04 -18.984429
POS Purchase 899 -4.831727e+05 -537.455693
Bill Payment 853 -4.276459e+05 -501.343389
--- TRANSACTION ACTIVITY BY AGE GROUP ---
count sum
AgeGroup
18-25 880 1.971767e+05
26-35 994 6.991372e+04
36-45 1085 -4.114068e+05
46-60 1397 -5.263432e+05
60+ 1644 -2.206116e+06
--- NET TRANSACTION VALUE BY OCCUPATION ---
Occupation
Business Owner 6.636860e+05
Student -1.620987e+05
Retired -9.069276e+05
Salaried -9.880232e+05
Self-Employed -1.482812e+06
Name: Amount, dtype: float64
--- BRANCH PERFORMANCE ---
TransactionCount TotalNetAmount
BranchName
Ahmedabad Main Branch 492 716963.743267
Jaipur Main Branch 553 637484.005361
Pune Main Branch 591 -84144.730000
Kochi Main Branch 504 -206312.423576
Chennai Main Branch 780 -367895.342576
Hyderabad Main Branch 528 -560903.951320
Mumbai Main Branch 632 -703871.660343
Kolkata Main Branch 573 -717591.109058